STA100 (Current Final) 2020

1. Draw Cumulative Frequency polygon from the given data...5 marks

2. write any two names of Statistical software packages. 2 marks.

Ans: These include SPSS, SAS, Statistica, Statgraph, Minitab, Stata, S-Plus, etc.

3. properties of sampling distribution X1 - X2 . 5 marks

Ans:

PROPERTIES OF THE SAMPLING DISTRIBUTION OF $\overline{X}_1 - \overline{X}_2$

Property No. 1:

The mean of the sampling distribution of $\overline{X}_1 - \overline{X}_2$, denoted by $\mu_{\overline{X}_1 - \overline{X}_2}$, is equal to the difference between population means, that is

$$\mu_{\overline{X}_1 - \overline{X}_2} = \mu_1 - \mu_2$$

Property No. 2:

In case of sampling with or without replacement from two *infinite* populations, the standard deviation of the sampling distribution of $\overline{X}_1 - \overline{X}_2$ (i.e. *standard error* of $\overline{X}_1 - \overline{X}_2$), denoted by $\sigma_{\overline{X}_1 - \overline{X}_2}$, is given by

$$\sigma_{\overline{X}_1 - \overline{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Property No. 3:

Shape of the distribution:

a) If the POPULATIONS are normally distributed, the sampling distribution of $\overline{X}_1 - \overline{X}_2$, regardless of sample sizes, will be *normal* with mean $\mu_1 - \mu_2$ and variance $\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$.

In other words, the variable

$$Z = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is normally distributed with zero mean and unit variance.

b) If the POPULATIONS are *non-normal* and if *both* sample sizes are *large*, (i.e., greater than or equal to 30), then the sampling distribution of the differences between means is approximately a *normal* distribution by the Central Limit Theorem.

In this case too, the variable

$$Z = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

will be approximately *normally* distributed with mean zero and variance one.

4. find the Mode - value from the given data. 2 marks

5. find the Mid-value of the given data. 2 marks

6. Write regression equation. 2 marks

$$Y = a + bX$$

7. Write Random samples... 5 marks

8. Disadvantages of range?

it has two serious disadvantages.
1. It *ignores* all the INFORMATION available from the intermediate observations.
2. It might give a MISLEADING picture of the spread in the data.

9. Types of measurement scale?

By *measurement*, we usually mean the assigning of number to observations or objects and scaling is a process of measuring.

The four scales of measurements are briefly mentioned below:

i. **NOMINAL SCALE:**
ii. **ORDINAL OR RANKING SCALE:**
iii. **INTERVAL SCALE:**
iv. **RATIO SCALE:**

10. Sampling distribution?

The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in a sample, etc.) is known as its sampling distribution

11. Primary and secondary data?

PRIMARY AND SECONDARY DATA:
Data that have been originally collected (raw data) and have not undergone any sort of statistical treatment, are called PRIMARY data. Data that have undergone any sort of treatment by statistical methods at least ONCE, i.e. the data that have been collected, classified, tabulated or presented in some form for a certain purpose, are called SECONDARY data.

12. Frequency polygon?

FREQUENCY POLYGON:
A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments.

13. Find covariance?

14. Regression?

15. Find mode

16. Set

A *set* is an *unordered* collection of *distinct* objects. Objects in the collection are called *elements* of the set.

17. experimental design

EXPERIMENTAL DESIGN
By an experimental design, we mean a plan used to collect the data relevant to the problem under study in such a way as to provide a basis for valid and objective inference about the stated problem. The plan usually includes:
• the selection of treatments whose effects are to be studied,
• the specification of the experimental layout, and
• the assignment of treatments to the experimental units.

18. scales

19. sample error

1. Sampling Error:
The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the parameter) is technically called the sampling error. For example,

Sampling error = $X - \mu$

Non-Sampling Error:
Besides sampling errors, there are certain errors which are not attributable to sampling but arise in the process of data collection, even if a complete count is carried out.

20. regression line

21. Mst measure

22. confidence difference upto 95%

23. Measure mean